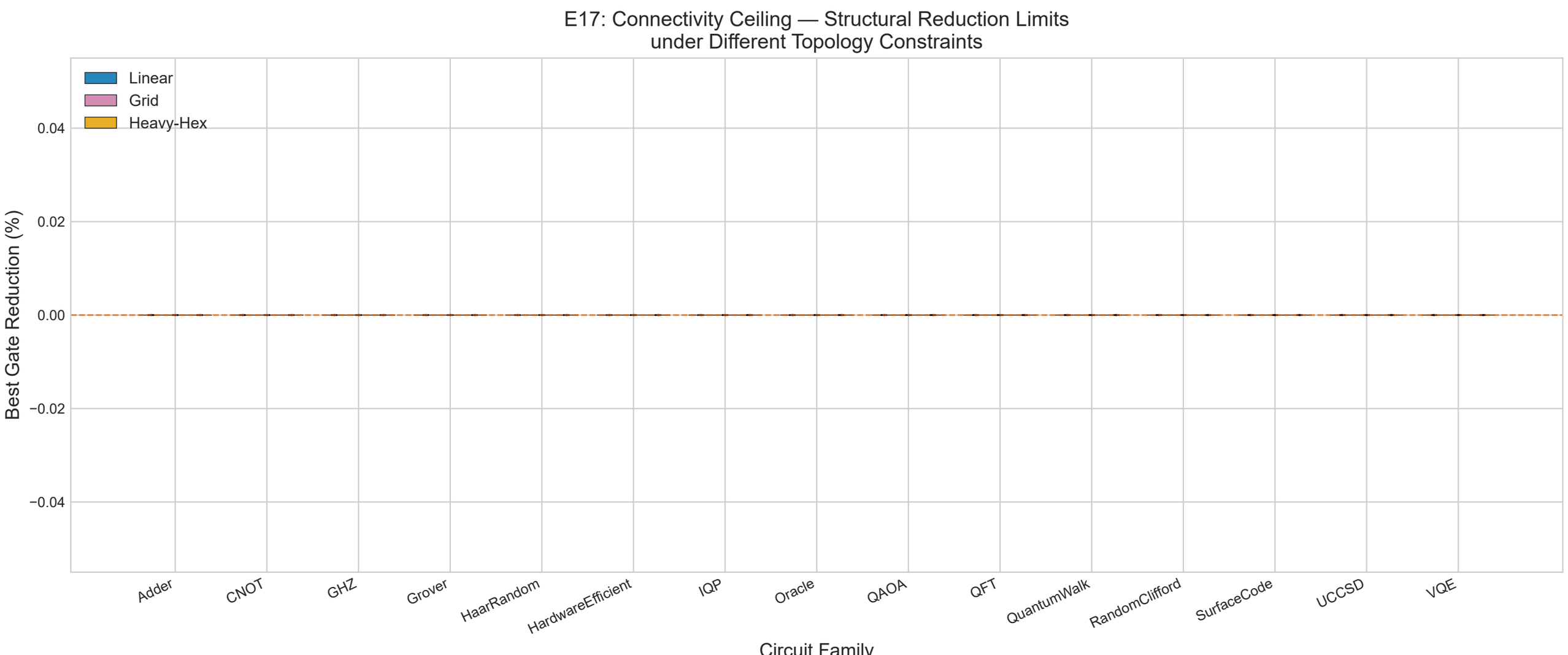
E17: Connectivity Ceiling — Structural Reduction Limits
under Different Topology Constraints
Linear
Grid
Heavy-Hex
Best Gate Reduction (%)
0.04
0.02
0.00
−0.02
−0.04
Adder
CNOT
GHZ
Grover
HaarRandom
HardwareEfficient
IQP
Oracle
QAOA
QFT
QuantumWalk
RandomClifford
SurfaceCode
UCCSD
VQE
Circuit Family